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Subject Code: DMITBA-01

Paper ID: 13441

GNIOT Institute of Management Studies – PGDM (Batch 2022-24)

(TRIMESTER – III) END TERM EXAMINATION, APRIL - 2023

STATISTICAL METHODS FOR DECISION MAKING USING R/PYTHON

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Write short notes on the following:**(10×3=30)**

- (a) Write about vectors in R
- (b) Write about logistic regression.
- (c) Explain different data structure in R
- (d) Explain about descriptive statistics? Write examples?
- (e) What is Box Plot? Explain importance of boxplot with example.
- (f) Find median and mode of following numbers
12,13,11,10,19,2,7,8,9,16,15,11
- (g) Explain the importance of dataframe?
- (h) What is the difference between a vector and a list in R?
- (i) How can you import data from a CSV file in R Studio? Provide an example.
- (j) What is the mean, median, and mode of a data set? How can you calculate these measures of central tendency in R?

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following:**(4×10=40)**

Q.2. (a) What is the difference between correlation and causation in statistics? How can you calculate correlation coefficients in R?

OR

(b) What is regression analysis, and how can it be used to model relationships between variables? Provide an example of regression analysis using R.

Q.3. (a) How can you create a scatter plot in R Studio? Provide an example.

OR

(b) Explain the concept of data wrangling in R. Provide an example of data wrangling using dplyr package.

Q.4. (a) How can you install packages in R Studio? Explain with an example.

OR

(b) How to create an empty matrix in R ?

Q.5. (a) Create a list containing a vector and a List.

OR

(b) Cluster Analysis in R

SECTION-C**Q.6. Attempt all parts of the following:****(2×15=30)**

6(a) Create a matrix taking a vector of numbers as input.

6(b) How to calculate Mean of several columns in Data Frame